

Threat Hunting: Living-off-the-Land Attacks on Windows

Overview

This project demonstrates **threat hunting and detection of Living-off-the-Land (LotL) attacks** in a Windows Active Directory environment using **Wazuh SIEM** and **Sysmon**.

Three attacker techniques were simulated from a Kali Linux machine against a domain-joined Windows host:

- **T1059.001** – PowerShell Execution
- **T1047** – Windows Management Instrumentation (WMI)
- **T1053.005** – Scheduled Task Persistence

All activity was logged, investigated, and mapped to the **MITRE ATT&CK framework**, following a SOC analyst workflow.

Lab Environment

Virtual Machines

- **Kali Linux** – Attacker
- **Windows Server** – Active Directory Domain Controller
- **Windows 10** – Domain-joined victim
- **Ubuntu Server** – Wazuh Manager

All systems are connected to the same internal virtual network.

Tools Used

Defensive

- Wazuh Agent
- Wazuh SIEM
- Sysmon (SwiftOnSecurity configuration)
- PowerShell Script Block Logging
- Windows Event Logging

Offensive

- Evil-WinRM
 - Impacket ([wmiexec.py](#))
 - Native Windows tools (PowerShell, CMD, [schtasks.exe](#))
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Data Sources

The following telemetry sources were used during investigations:

- Sysmon (Process Create, File Create)
- Windows Security and Operational Logs
- PowerShell Script Block Logging
- Wazuh SIEM alerts and MITRE mappings

Key Wazuh fields analyzed include:

- `data.win.system.eventID`
 - `data.win.eventdata.Image`
 - `data.win.eventdata.CommandLine`
 - `rule.mitre.id`
-

Attack Simulations and Detection

1. PowerShell Execution (T1059.001)

Attack Execution

Executed via Evil-WinRM using an encoded PowerShell command:

```
powershell -enc  
ZQBjAGgAbwAgACIAQQBUAFQAQQBDAEsAXwBUAEUAUwBUACIAIAA+ACAAQwA6AFwAVwBpAG  
4AZABvAHcAcwBcAFQAZQBtAHAAXABhAHQAdABhAGMAawAuAHQAdAA=
```

Detection Evidence

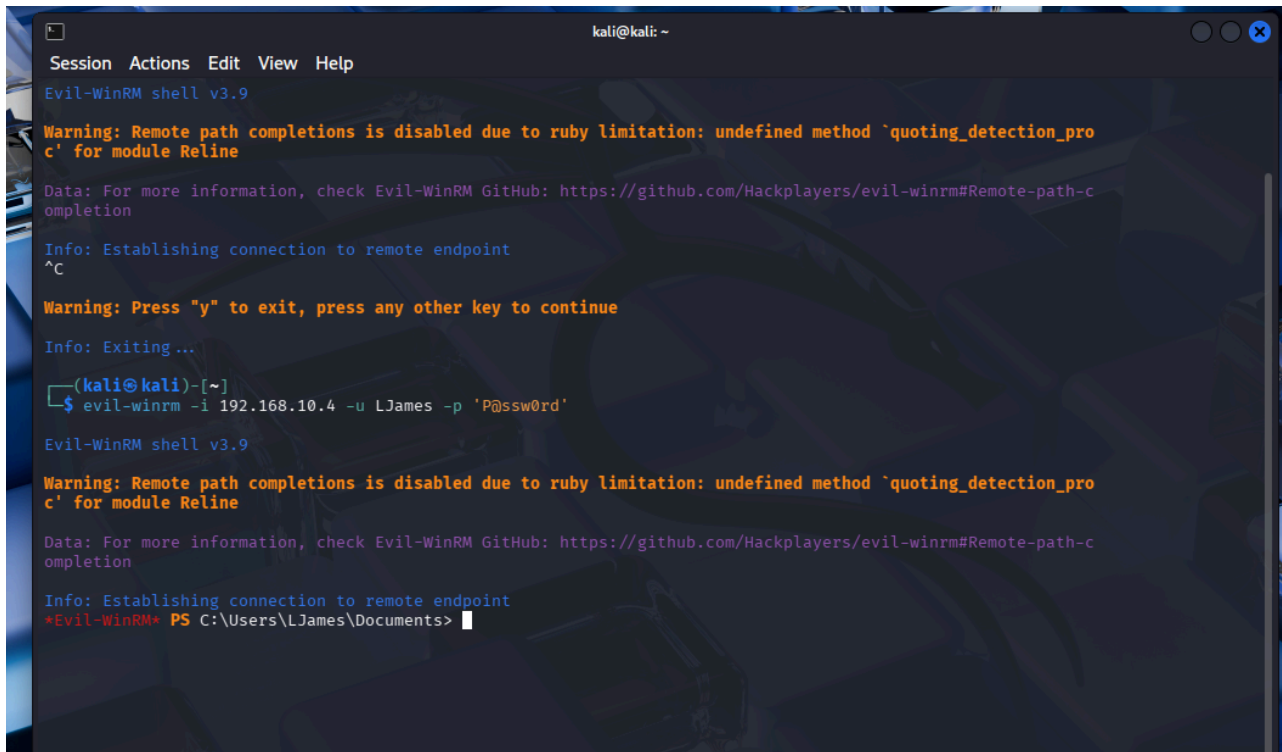
Sysmon

- Event ID: 11 (File Create)
- Process: `powershell.exe`
- File created: `__PSScriptPolicyTest_*.ps1`

Wazuh

- MITRE Technique: **T1059.001**
 - PowerShell execution detected and correlated
-

Screenshots



```
kali@kali: ~  
Session Actions Edit View Help  
Evil-WinRM shell v3.9  
  
Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_pro  
c' for module Reline  
  
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-c  
ompletion  
  
Info: Establishing connection to remote endpoint  
^C  
  
Warning: Press "y" to exit, press any other key to continue  
  
Info: Exiting...  
(kali@kali)-[~]  
$ evil-winrm -i 192.168.10.4 -u LJAMES -p 'P@ssw0rd'  
  
Evil-WinRM shell v3.9  
  
Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_pro  
c' for module Reline  
  
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-c  
ompletion  
  
Info: Establishing connection to remote endpoint  
*Evil-WinRM* PS C:\Users\LJAMES\Documents>
```

Operational Number of events: 59,431 (!) New events available

Level	Date and Time	Source	Event ID	Task Category
 Information	2/4/2026 7:11:39 AM	Sysmon	13	Registry value set (rule: RegistryEvent)
 Information	2/4/2026 7:11:39 AM	Sysmon	13	Registry value set (rule: RegistryEvent)
 Information	2/4/2026 7:11:39 AM	Sysmon	7	Image loaded (rule: ImageLoad)
 Information	2/4/2026 7:08:31 AM	Sysmon	7	Image loaded (rule: ImageLoad)
 Information	2/4/2026 7:07:47 AM	Sysmon	7	Image loaded (rule: ImageLoad)
 Information	2/4/2026 7:07:35 AM	Sysmon	7	Image loaded (rule: ImageLoad)
 Information	2/4/2026 7:07:33 AM	Sysmon	10	Process accessed (rule: ProcessAccess)
 Information	2/4/2026 7:07:33 AM	Sysmon	22	Dns query (rule: DnsQuery)
 Information	2/4/2026 7:07:33 AM	Sysmon	17	Pipe Created (rule: PipeEvent)
 Information	2/4/2026 7:07:32 AM	Sysmon	26	File Delete logged (rule: FileDeleteDet...)
 Information	2/4/2026 7:07:32 AM	Sysmon	11	File created (rule: FileCreate)
 Information	2/4/2026 7:07:32 AM	Sysmon	7	Image loaded (rule: ImageLoad)

Event 11, Sysmon

General Details

☒ Friendly View ☐ XML View

+ System

- EventData

RuleName technique_id=T1059.001,technique_name=PowerShell

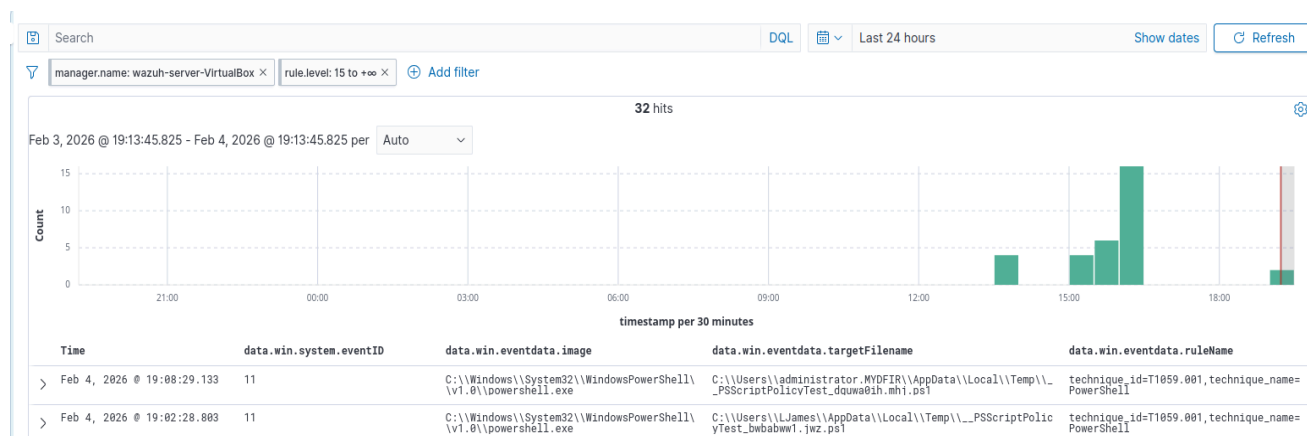
UtcTime 2026-02-04 15:07:32.422

ProcessGuid {6d651f2b-60b3-6983-2a01-000000000a00}

ProcessId 6328

Image C:\Windows\System32\WindowsPowerShell\v1.0
\powershell.exe

TargetFilename C:\Users\administrator.MYDFIR\AppData\Local\Temp\



Analysis

Encoded PowerShell execution is a common LotL technique. Script Block Logging and Sysmon successfully captured execution artifacts, which were correlated in Wazuh and mapped to MITRE ATT&CK.

2. WMI Remote Execution (T1047)

Attack Execution

Executed from Kali Linux using Impacket:

```
python3 wmiexec.py mydfir/LJames:'P@ssw0rd'@192.168.10.4
```

Remote command executed on the Windows host:

```
cmd.exe /c echo WMI_ATTACK > C:\\Windows\\Temp\\wmi.txt
```

Detection Evidence

Sysmon

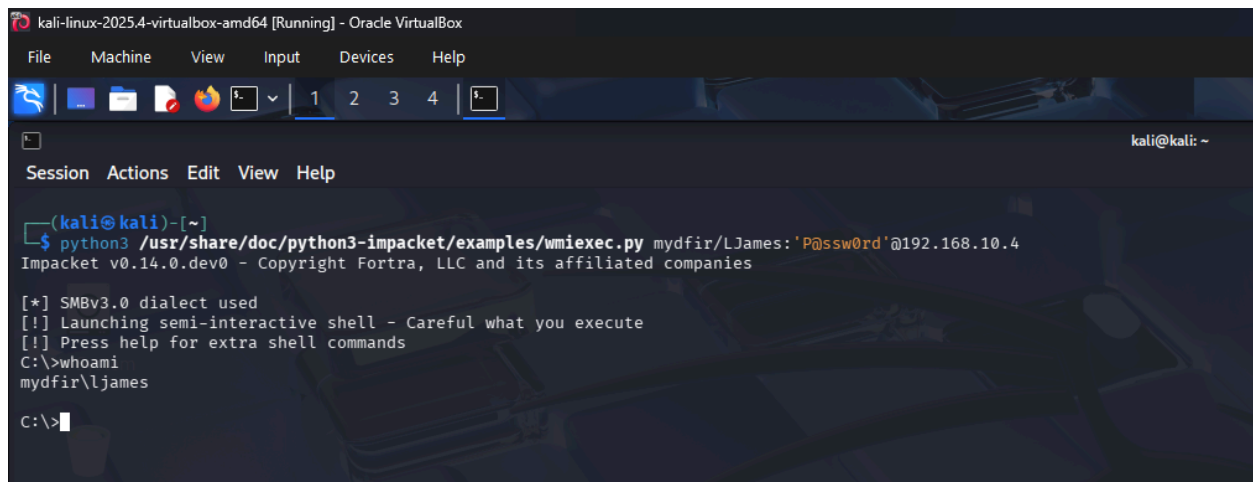
- Event ID: 1 (Process Create)

- Process: `cmd.exe`
- Parent command line shows WMI-based execution

Wazuh

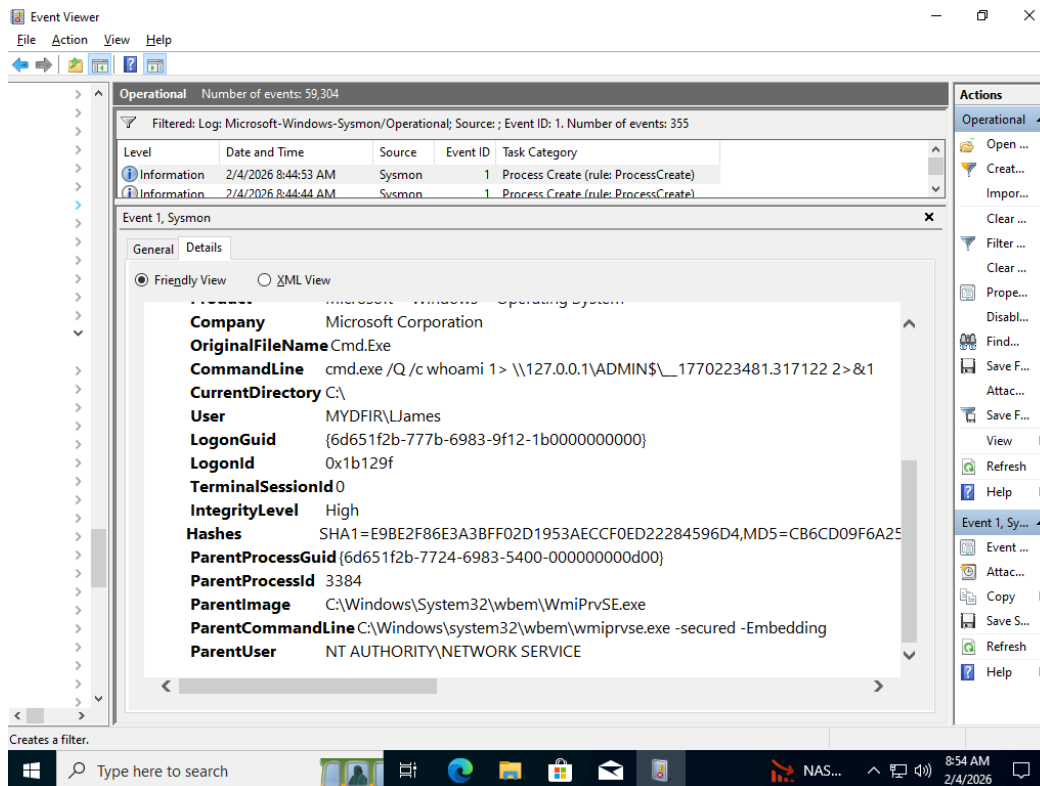
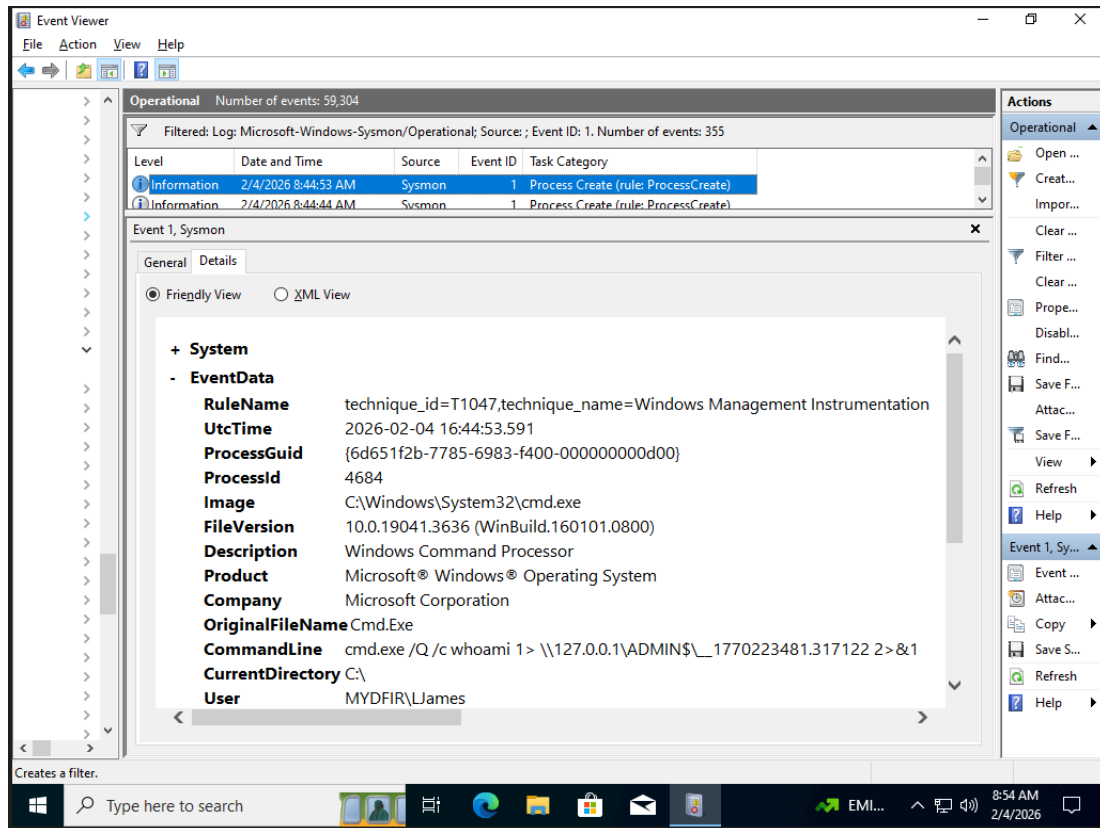
- MITRE Technique: `T1047`

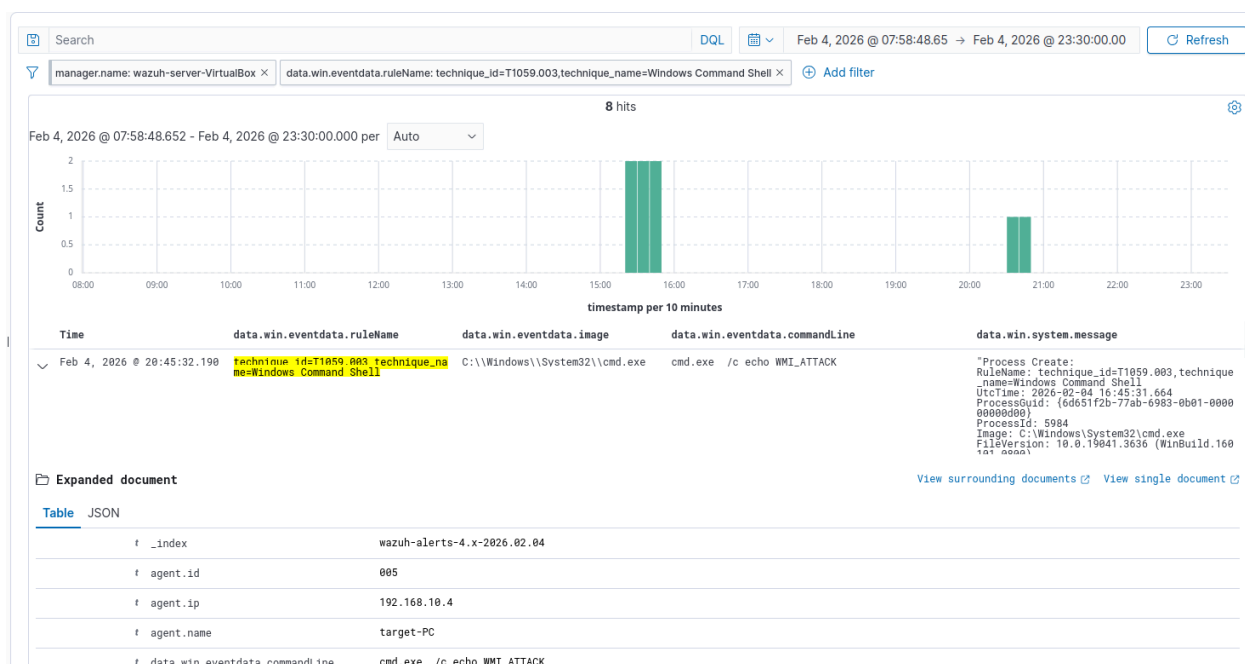
Screenshots



The screenshot shows a Kali Linux terminal window titled "kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VirtualBox". The terminal displays the execution of the `python3 /usr/share/doc/python3-impacket/examples/wmiexec.py` script with the arguments `mydfir/LJames:'P@ssw0rd'@192.168.10.4`. The output shows the script connecting to the remote host and launching a semi-interactive shell. The user `mydfir\Ljames` is prompted to enter a command, and the user enters `whoami`, resulting in the output `mydfir\Ljames`.

```
kali@kali: ~  
$ python3 /usr/share/doc/python3-impacket/examples/wmiexec.py mydfir/LJames:'P@ssw0rd'@192.168.10.4  
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies  
[*] SMBv3.0 dialect used  
[!] Launching semi-interactive shell - Careful what you execute  
[!] Press help for extra shell commands  
C:\>whoami  
mydfir\Ljames  
C:\>
```





Analysis

WMI enables stealthy remote command execution without interactive login. Sysmon revealed non-interactive process creation, and Wazuh correlated this activity to the appropriate MITRE technique.

3. Scheduled Task Persistence (T1053.005)

Attack Execution

Executed via Evil-WinRM:

```
schtasks /create /sc minute /mo 5 /tn "WindowsUpdateCheck" /tr  
"cmd.exe /c echo PERSISTENCE > C:\\Windows\\Temp\\persist.txt" /ru SYSTEM
```

Detection Evidence

Sysmon

- Event ID: 1 (Process Create)
- Process: `schtasks.exe`
- Full task creation command logged

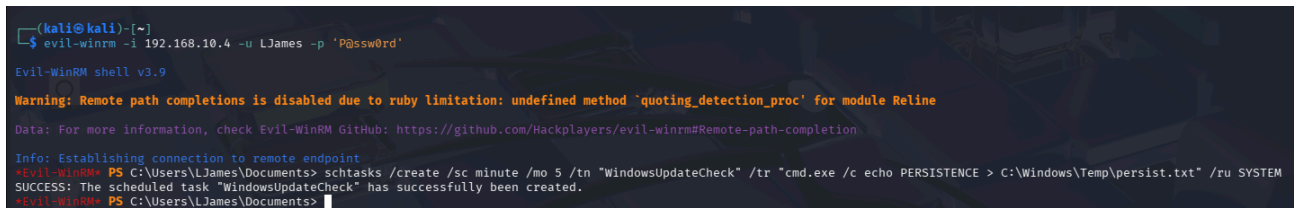
Wazuh

- MITRE Technique: `T1053`

Persistence Artifact

- File created: `C:\Windows\Temp\persist.txt`

Screenshots

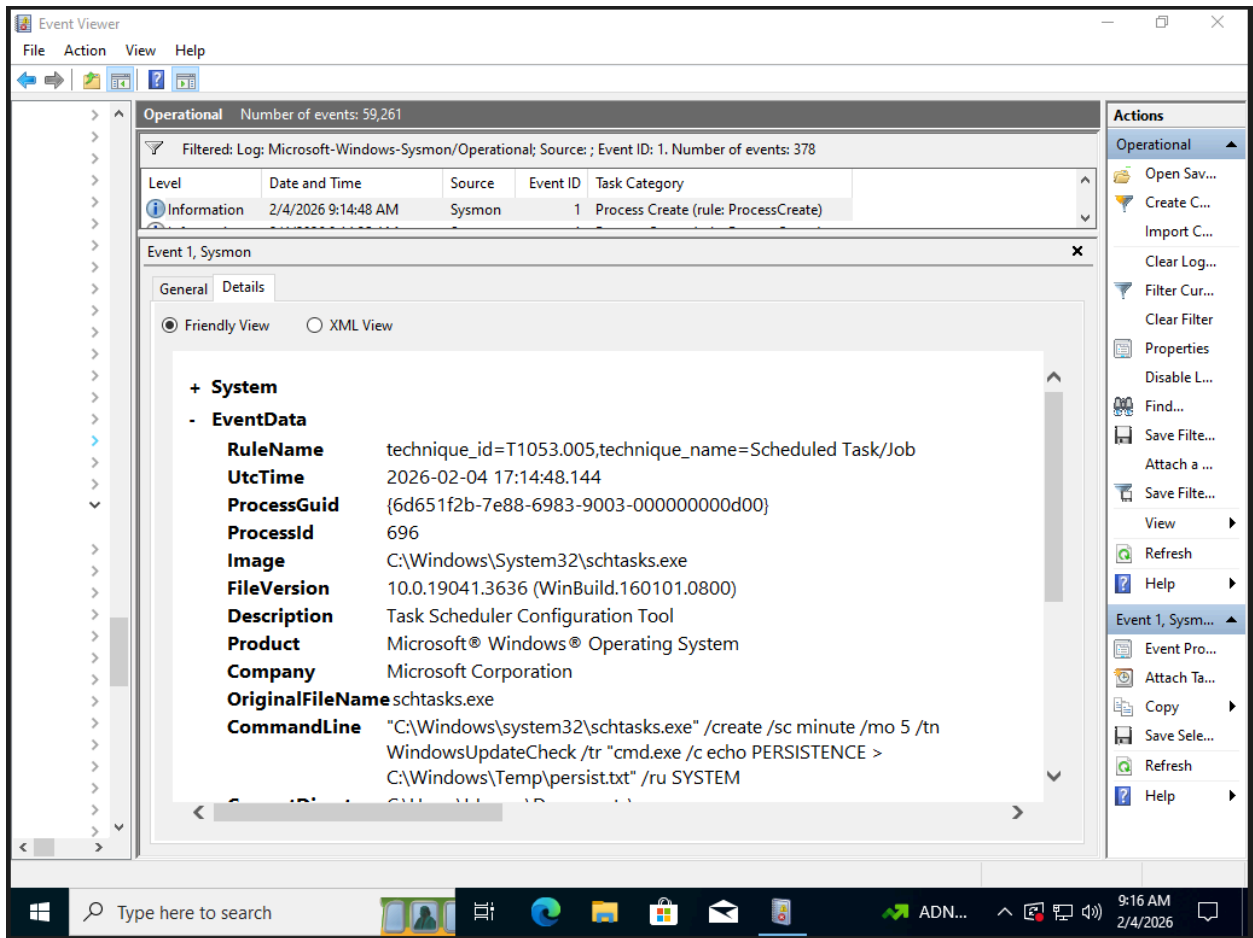


```
(kali@kali)-[~]
└─$ evil-winrm -i 192.168.10.4 -u LJAMES -p 'P@ssw0rd'
Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\LJames\Documents> schtasks /create /sc minute /mo 5 /tn "WindowsUpdateCheck" /tr "cmd.exe /c echo PERSISTENCE > C:\Windows\Temp\persist.txt" /ru SYSTEM
SUCCESS: The scheduled task "WindowsUpdateCheck" has successfully been created.
*Evil-WinRM* PS C:\Users\LJames\Documents>
```



Time	data.win.system.eventID	data.win.eventdata.ruleName	data.win.eventdata.image
Feb 4, 2026 @ 21:15:09.628	7	technique_id=T1053,technique_name=Scheduled Task	C:\Windows\System32\schtasks.exe

Expanded document

Table JSON

_index	wazuh-alerts-4.x-2026.02.04
agent.id	005
agent.ip	192.168.10.4
agent.name	target-PC
data.win.eventdata.company	Microsoft Corporation
data.win.eventdata.description	Task Scheduler COM API
data.win.eventdata.fileVersion	10.0.19041.3636 (WinBuild.160101.0800)
data.win.eventdata.image	C:\Windows\System32\schtasks.exe

Analysis

Scheduled tasks are commonly abused for persistence. Sysmon captured task creation and execution, and Wazuh correctly mapped the activity to T1053.

MITRE ATT&CK Coverage

This project covers the following techniques:

- **T1059.001** – PowerShell
 - **T1047** – Windows Management Instrumentation
 - **T1053.005** – Scheduled Task / Job
-

Key Takeaways

- Living-off-the-Land techniques are highly detectable with proper endpoint logging
 - Sysmon significantly improves visibility into attacker behavior
 - Wazuh effectively correlates endpoint telemetry and maps to MITRE ATT&CK
 - Persistence mechanisms leave reliable forensic artifacts
-

Conclusion

This project demonstrates a **realistic SOC threat-hunting workflow**, including:

- Attack simulation
- Endpoint telemetry collection
- SIEM investigation
- MITRE ATT&CK mapping
- Evidence documentation

It reflects practical, job-ready skills expected of a **SOC Analyst / DFIR Intern**.

Disclaimer

This project was conducted in a **controlled lab environment for educational purposes only**.